THE EFFECT OF 2% *CHLORELLA VULGARIS* SUPPLEMENTATION IN BROILER FEED ON THE QUALITY AND SENSORY CHARACTERISTICS OF CHICKEN HAMS

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Introduction

Modern poultry production seeks innovative ways to enhance the nutritional value of meat products. Microalgae, specifically *Chlorella vulgaris*, are considered a "superfood" due to their high content of essential amino acids, polyunsaturated fatty acids, and natural pigments. This study investigates how a 2% addition of Chlorella to the feed mixture of broiler chickens (Cobb 500) influences the final quality parameters of processed chicken hams. The focus is on whether the nutritional benefits of algae can be transferred to the product without compromising its technological and sensory appeal.

Materials and methodology

- Animals & Feed: Broilers (n=24); Control vs. Experimental (2% *Chlorella vulgaris* in BR2).
- Product: Chicken hams (breast/thigh) – MeatLab MENDELU.
- Chemical (ISO): Dry matter, Protein (Kjeldahl), Fat (Soxhlet), Salt (Volhard).
- Texture: Cutting force (N), MORS test on TIRA test 27025.
- Color: CIELAB via CM-3500d (Konica Minolta).
- Sensory: Trained panel (n=10), 9-point scale.
- Descriptors: Appearance, Colour, Aroma, Texture, Taste, Overall Acceptance.

Results and discussion

A. Chemical Composition (Nutritional Value)

The addition of Chlorella significantly improved the protein content in both types of ham. This suggests that algae-derived nutrients are efficiently utilized by the chickens and deposited into the muscle tissue.

Chemical composition of chicken hams

Parameter	Control (Breast)	Chlorella (Breast)	Control (Thigh)	Chlorella (Thigh)
Dry matter (%)	22,85±0,04	**25,33±0,00**	24,18±0,03	23,94±0,10
Proteins (%)	19,67±0,27	**21,79±0,19**	16,68±0,30	18,34±0,05
Fat (%)	1,32±0,06	1,38±0,16	5,31±0,22	**3,25±0,16**
Salt (%)	1,85±0,00	1,90±0,02	1,75±0,02	1,81±0,02

B. Texture and Physical Properties

Measurements using the MORS test revealed that the experimental group (Chlorella) had lower cutting force values for breast hams (3,73 N) compared to the control group (4,72 N). This indicates that the addition of algae may contribute to a more tender texture in the final product.

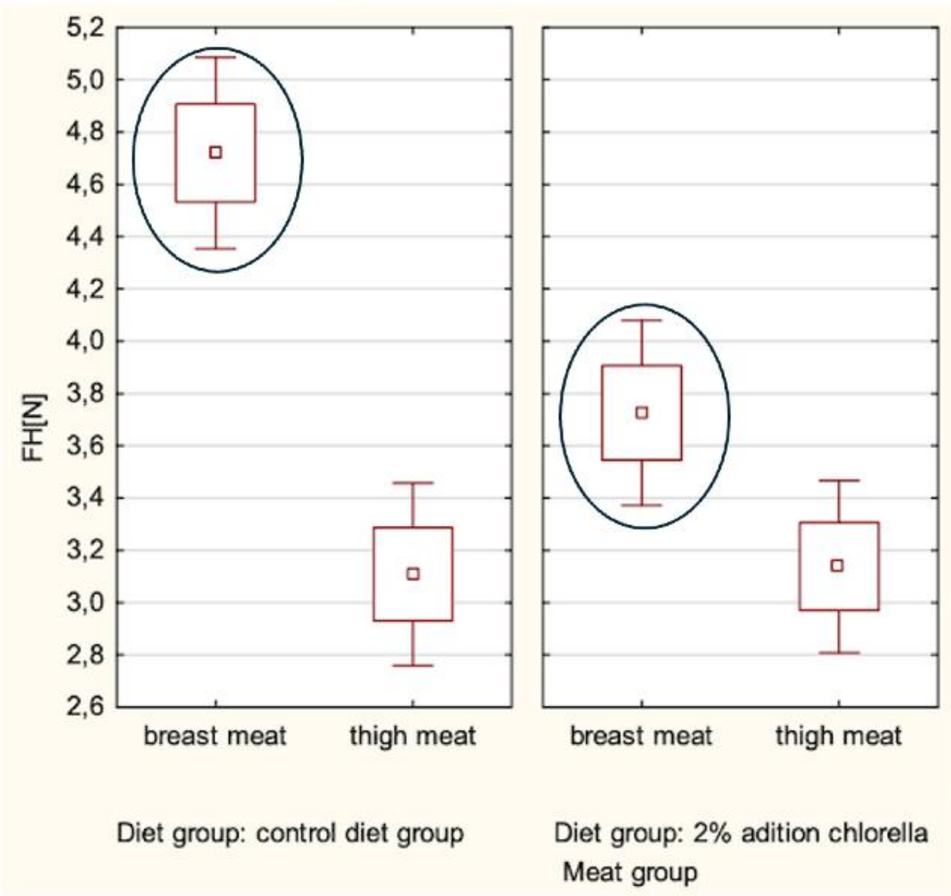

Figure 1: Comparison of FH [N] values between breast and thigh meat in the control group and the 2% chlorella addition group.

C. Colour and Sensory Acceptance

The most significant impact of Chlorella was observed in the colour profile. The yellowness (value) increased significantly in the Chlorella hams (13,47 vs. 11,62 in control). In the sensory evaluation, this led to a lower score for appearance, as consumers are not accustomed to yellowish tints in poultry ham.

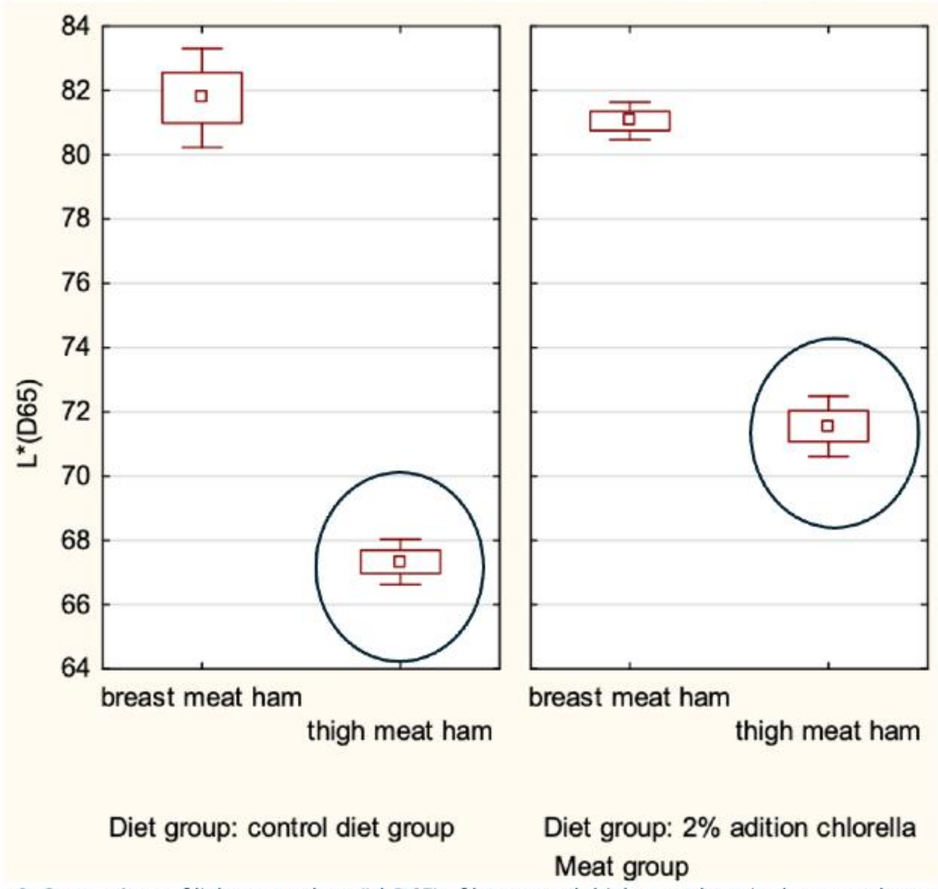

Figure 2: Comparison of lightness values (L* D65) of breast and thigh meat ham in the control group and the 2% chlorella addition group.

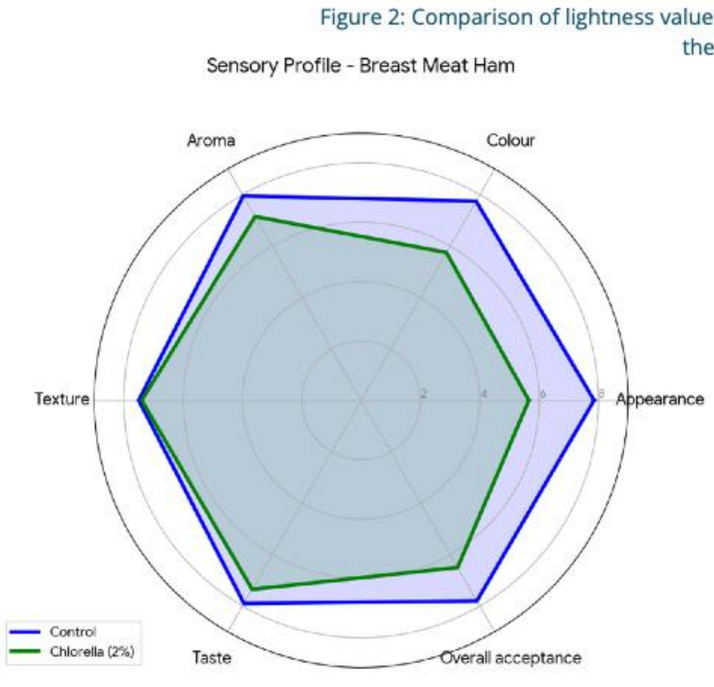

Figure 3: Sensory profile of chicken breast hams comparing the control group and the group with 2% Chlorella addition

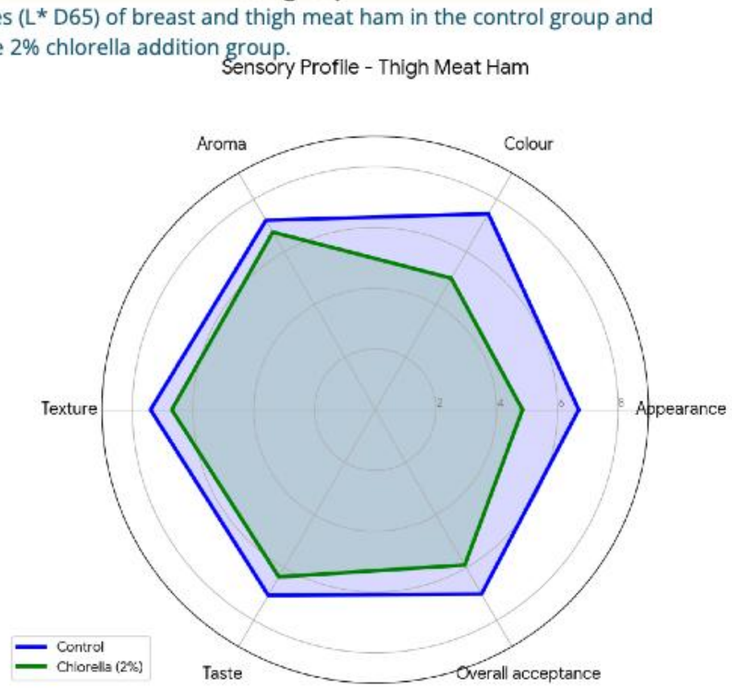

Figure 4: Sensory profile of chicken thigh hams comparing the control group and the group with 2% Chlorella addition

Conclusion

- **Appearance & Colour:** Lower sensory scores due to green pigmentation; however, instrumental lightness (L*)remained **high**, preventing a dull or dark appearance.
- **Nutritional Profile:** Significantly **higher** protein content in Chlorella hams compared to the control group.
- **Tenderness (MORS): Lower shear force** values, resulting in more tender and softer meat texture.
- **Taste & Aroma: Comparable** to the control group, with **high acceptance maintained** (especially in breast meat).
- **Meat Type: Breast meat** hams achieved **better overall performance** (higher proteins and sensory scores) than thigh meat hams.
- **Overall Impact:** Chlorella **improves nutritional value** and **tenderness** while preserving **taste quality,** despite the change in color.

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